

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – REPRODUCTION IN PLANTS (SUB-STRAND 2.1)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Your task is to answer the driving question: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit? Write a Claim-Evidence-Reasoning (CER) explanation and include a revised, annotated diagram of crop reproduction. Use evidence from our 8 lessons, including your flower and fruit practical lab. Refer to Kenyan crops such as cassava, sugarcane, sweet potato, maize, beans, pawpaw and avocado. Write in your own words, use correct biological terms, and label your diagram clearly. Below, each section has a prompt and a model example (exemplar) to show what a strong answer looks like. Do not copy the exemplars. Use them to learn how to build your own explanation.

Part 1: Claim – Answering the Driving Question

State your claim in 2-4 sentences. Explain that plants use two reproductive strategies, and say which strategy cassava, sugarcane, sweet potato, maize, beans, pawpaw and avocado depend on. Mention why this matters to a farmer.

Plants reproduce in two main ways: asexual and sexual reproduction. Cassava, sugarcane and sweet potato can grow from pieces of stem or vine (cuttings) because their vegetative parts can grow into whole new plants without seeds or partners. Maize and beans, however, are grown from seeds, which come from sexual reproduction, where male and female gametes join. Pawpaw and avocado trees often need another tree nearby because pollen must reach the flowers of a different, compatible plant before fruit can form. Farmers use asexual reproduction when they want fast, identical plants and sexual reproduction when they want seeds and genetic variety.

Part 2: Evidence – Asexual Reproduction (Plants Without Seeds)

Give at least three pieces of evidence from our lessons about how plants reproduce without seeds. Include the vegetative structures (e.g. tubers, bulbs, rhizomes,

Evidence 1: When a farmer in Kisumu plants a cassava or sugarcane stem cutting, buds on the stem grow roots and shoots and become a whole new plant. No flower, seed or partner is needed. Evidence 2: Plants have special vegetative structures for this: sweet potato vines and tubers, runners on creeping grasses, rhizomes in ginger and arrowroot, and bulbs in onions. Each structure stores food and has buds that can grow into a new plant. Evidence 3: The new plants form by mitosis, a type of cell division that copies the parent's cells exactly. In our observations, offspring from cuttings looked the same as the parent, because they are genetically identical. This is why a farmer in Kericho or Nairobi can quickly plant a whole field of uniform sweet potatoes from a few

runners, stem cuttings), examples from Kenyan farms, and what happens to the cells and the genes of the offspring.	vines.
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Part 3: Evidence – Sexual Reproduction, Pollination and Fertilisation

Use evidence from the flower dissection and fruit practical to explain how seeds and fruits form. Describe flower parts (stamen, anther, stigma, style, ovary, ovule), the difference between self- and cross-pollination, how pollen tube growth leads to double fertilisation, and what the ovule and ovary become afterwards. Explain why a single pawpaw or avocado tree may flower heavily but not set fruit.	In our flower dissection, we found that the anther (male part) produces pollen containing male gametes, while the ovary holds ovules containing the egg cell (female part). Pollination happens when pollen lands on the stigma. Many plants favour cross-pollination (pollen from another plant) over self-pollination. They do this through structural barriers, different timing of male and female parts maturing (protandry or protogyny), or self-incompatibility, where a plant rejects its own pollen. This explains the lone pawpaw or avocado tree covered in flowers with no fruit: its own pollen cannot fertilise its own flowers, or its male and female parts are ready at different times, so it needs a partner tree nearby. After a compatible pollen grain lands on the stigma, a pollen tube grows down the style into the ovule and delivers two male gametes. In double fertilisation, one sperm nucleus fuses with the egg cell to form the diploid zygote ($2n$), which grows into the embryo. The other fuses with two polar nuclei to form the triploid endosperm ($3n$), the food store, like the white starchy part of a maize grain that we eat as ugali. Afterwards, the ovules become seeds and the ovary wall becomes the fruit. In our fruit practical, we cut open fruits such as beans pods and passion fruit and saw the seeds attached in patterns (placentation), such as along the edge (marginal) in bean pods.
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Part 4: Reasoning – Why Two Strategies Exist

Link your evidence to your claim. Explain the advantages and disadvantages of asexual and sexual reproduction, and why cross-pollination and seed production help plant populations survive changes such as drought, disease or pests. Give a Kenyan example.	Asexual reproduction is fast and reliable because it needs no pollinators or partner, and it copies a successful plant exactly. This is why farmers use cassava cuttings: they get the same high-yielding plant each time. The disadvantage is that all the plants are genetically identical, so if a disease like cassava mosaic disease attacks, all of them can be harmed. Sexual reproduction mixes genes from two parents through gametes, so seeds produce offspring that differ from each other. This variation means that in a changing environment, such as a drought in the Rift Valley or a new maize pest, some plants will probably survive and pass on their genes. Cross-pollination increases this variation even more than self-pollination, which is why mechanisms like self-incompatibility exist. So the reason some crops grow from cuttings and others need seeds and partner trees is that each method solves a different problem: asexual reproduction gives speed and uniformity, while sexual reproduction gives variety, adaptability and seeds that can be stored and spread.
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Part 5: Revised Annotated Diagram

Draw a final diagram with two linked pathways: (A) asexual reproduction in a Kenyan crop such as cassava or sweet potato, and (B) sexual reproduction from pollination to fruit and seed formation in a	Pathway A (cassava): Parent plant → stem cutting with buds → roots and shoots grow by mitosis → new plant identical to parent. Notes: 'No seed or flower needed. Genetically identical offspring.' Pathway B (pawpaw/avocado): Flower of tree 1 with anther → pollen carried by insects or wind → lands on stigma of flower on tree 2 (cross-pollination) → pollen tube grows down the style → two male gametes enter the ovule → double fertilisation: sperm + egg = zygote ($2n$); sperm + two polar nuclei = endosperm ($3n$) → ovule becomes seed; ovary becomes fruit. Notes: 'Lone tree = no compatible pollen, so no fruit.' Revision note: 'At first I thought flowers alone made fruit. Now I know fertilisation by compatible pollen must happen first.'
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crop such as pawpaw, avocado or beans. Label all key parts and processes and add short notes explaining what is happening at each step. Add one note showing how your ideas changed since your first model.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the driving question	Claim clearly and accurately answers the driving question, names both asexual and sexual strategies, and links them to specific Kenyan crops.	Claim answers the question and names both strategies, but links to crops are limited or partly unclear.	Claim is missing, incomplete or does not clearly address why crops reproduce differently.
Evidence from lessons and practical	Uses several accurate pieces of evidence from lessons and the flower/fruit practical, covering vegetative structures, pollination, double fertilisation and fruit/seed formation.	Uses some accurate evidence, but one area (e.g. double fertilisation or placentation) is brief or lacks practical observations.	Evidence is minimal, inaccurate or not connected to class investigations.
Scientific reasoning	Explains clearly why a lone tree may not fruit, compares mitosis-based identical offspring with gamete-based variation, and links variation to survival in changing conditions.	Explains most connections between evidence and claim, but some reasoning about variation or pollination barriers is incomplete.	Mostly describes facts without linking them to the claim, or contains major misconceptions.
Annotated diagram	Diagram shows both pathways accurately, with correct labels, process notes (mitosis, pollen tube, double fertilisation, $2n/3n$) and a note on how thinking changed.	Diagram shows both pathways with mostly correct labels, but some notes or processes are missing.	Diagram is incomplete, has several incorrect or missing labels, or shows only one pathway.
Scientific vocabulary and communication	Uses correct terms precisely, writes in clear organised paragraphs, and uses Kenyan examples effectively.	Uses mostly correct terms and a clear structure with occasional errors.	Uses few scientific terms or the writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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